Brain Tumor Detection and Analysis Using Deep Learning, Radiomics, and Explainable Artificial Intelligence

1. Project Overview

The proposed project presents an intelligent computer-aided diagnosis framework for automated brain tumor analysis from Magnetic Resonance Imaging (MRI) scans using deep learning and explainable artificial intelligence. Brain tumors are among the most critical neurological disorders, where early and accurate identification plays a significant role in treatment planning and improving patient survival. Manual examination of MRI scans is often time-consuming and depends heavily on the expertise of radiologists. Therefore, there is an increasing need for automated systems capable of accurately identifying abnormal brain tissues while providing interpretable results.

The developed system integrates multiple artificial intelligence techniques into a unified framework. Initially, brain MRI images are uploaded through a user-friendly Streamlit web interface, where they undergo image preprocessing to standardize their dimensions and intensity values. The preprocessed images are then analyzed using a Dynamic U-Net segmentation network, which performs pixel-level tumor segmentation by identifying abnormal tissue regions. Unlike conventional classification systems that merely indicate the presence or absence of a tumor, the segmentation network accurately localizes the tumor boundary, thereby assisting clinicians in understanding the exact extent of the abnormal region.

To improve the transparency of the prediction process, the framework incorporates a Gradient-weighted Class Activation Mapping (Grad-CAM) module. Grad-CAM highlights the image regions that contribute most significantly to the deep learning model's prediction, allowing clinicians to visually verify whether the model is focusing on clinically relevant structures. In addition to segmentation and visualization, the project also performs radiomic feature extraction from the segmented tumor region. Radiomic features quantitatively describe tumor characteristics such as size, shape, intensity distribution, and morphological properties, providing valuable biomarkers that may support future clinical analysis or predictive modeling.

The entire framework has been implemented using Python with PyTorch as the deep learning framework, OpenCV and TorchVision for image processing, and Streamlit for interactive visualization. The integration of deep learning segmentation, explainable AI, radiomics, and an intuitive graphical interface makes the proposed system a comprehensive solution for intelligent brain tumor analysis.

2. Working Principle

The proposed framework follows a sequential image analysis pipeline beginning with MRI image acquisition and ending with visualization of the segmented tumor and explainability results. Initially, the user uploads a brain MRI image through the Streamlit-based graphical interface. The uploaded image is accepted in common image formats such as PNG, JPG, or JPEG and is immediately validated before further processing.

During preprocessing, the image is converted into grayscale and resized to a fixed resolution of 512 × 512 pixels to maintain consistency with the dimensions used during model training.

The image is then normalized and transformed into a tensor representation suitable for deep learning inference using the PyTorch framework.

The standardized image is subsequently provided as input to the Dynamic U-Net segmentation model. The encoder portion of the network extracts hierarchical spatial features from the MRI image through multiple convolutional operations, whereas the decoder reconstructs these features to produce a dense pixel-wise segmentation map. The predicted probability map is thresholded to generate a binary mask representing the tumor region. This binary mask is overlaid on the original MRI image using a red color mask to improve visual interpretation.

Following segmentation, the Grad-CAM module computes activation maps using the gradients of the final convolutional layer. These activation maps identify the image regions that contribute most strongly to the segmentation output, thereby increasing the interpretability of the deep learning model. The Grad-CAM heatmap is blended with the original MRI scan and displayed alongside the segmentation results.

Finally, radiomic features are extracted from the segmented tumor region. These features numerically describe the tumor's geometric and intensity characteristics, including voxel count, volume, elongation, flatness, axis lengths, and pixel intensity statistics. All generated outputs, including the original MRI image, segmented tumor, Grad-CAM visualization, and radiomic measurements, are presented through the Streamlit interface for user interaction and analysis.

3. Proposed System Architecture

The proposed system architecture consists of six major functional layers that work together to perform automated brain tumor analysis. The first layer is the user interface, implemented using Streamlit, through which users upload MRI images and visualize the generated results. This interface acts as the communication layer between the user and the artificial intelligence framework.

The uploaded MRI image is forwarded to the preprocessing layer, where image standardization is performed. During this stage, the image is converted into grayscale, resized to a resolution of 512 × 512 pixels, normalized, and converted into tensor format. These operations ensure that every MRI image possesses identical characteristics before being processed by the deep learning models.

The standardized image is then supplied to the Dynamic U-Net segmentation model. This model consists of an encoder-decoder architecture capable of learning complex spatial representations of brain tissues. The encoder gradually extracts low-level and high-level image features through convolutional operations, while the decoder reconstructs these features into a pixel-wise segmentation mask representing the tumor region. A threshold operation is applied to obtain the final binary segmentation output.

Once segmentation is completed, two parallel processing stages are performed. The first stage executes Grad-CAM to generate explainable visualizations by computing gradient-based activation maps from the trained network. These heatmaps highlight the regions that most strongly influence the model's prediction, thereby improving interpretability and clinical confidence.

The second stage performs radiomic feature extraction using the segmented tumor mask. Various quantitative descriptors are computed, including voxel volume, elongation, flatness, major and minor axis lengths, least axis length, mean intensity, maximum intensity, and minimum intensity. These radiomic features provide objective numerical information regarding tumor morphology and tissue characteristics.

Finally, all generated information is combined within the visualization layer. The system simultaneously displays the original MRI image, segmented tumor overlay, Grad-CAM heatmap, and extracted radiomic features through the Streamlit dashboard, providing clinicians with a comprehensive view of the analysis.

4. Proposed Methodology

The proposed methodology is organized into five sequential stages, beginning with MRI acquisition and ending with visualization of the analyzed results.

Initially, MRI images are uploaded through the web interface. Since MRI images originate from different scanners and imaging protocols, preprocessing is necessary to maintain consistency across all inputs. Each uploaded image is converted into grayscale, resized to 512 × 512 pixels, normalized, and transformed into a tensor representation suitable for deep learning inference.

Following preprocessing, the standardized image is processed by the Dynamic U-Net segmentation model. The encoder extracts hierarchical image features through successive convolutional layers and pooling operations, whereas the decoder progressively reconstructs these features to produce a detailed segmentation map. A threshold value of 0.5 is applied to convert the predicted probability map into a binary tumor mask. The segmented region is subsequently highlighted in red and superimposed on the original MRI image for improved visualization.

To improve the transparency of the deep learning prediction, the framework integrates Grad-CAM explainability. The gradients of the final convolutional layer are propagated backward to compute class activation maps. These activation maps reveal the image regions contributing most significantly to the segmentation output. The generated heatmap is normalized, colorized using the JET color map, and blended with the original MRI image to provide an interpretable visualization of the network's attention.

The segmented tumor mask is then used to compute radiomic features. The extracted features include first-order intensity statistics such as mean, minimum, and maximum intensity values, together with geometric descriptors including elongation, flatness, least axis length, major axis length, minor axis length, voxel count, and tumor volume. These features quantitatively characterize tumor morphology and can be utilized for future predictive modeling or clinical assessment.

Finally, all generated outputs are displayed through the Streamlit interface. The dashboard enables clinicians to simultaneously examine the original MRI image, segmented tumor region, Grad-CAM visualization, and quantitative radiomic measurements, thereby providing both visual and numerical evidence for clinical decision support.

5. Module Description

The proposed system is divided into several interconnected functional modules, each responsible for a specific stage of the analysis pipeline.

The Image Acquisition Module accepts MRI images uploaded by the user through the Streamlit interface. This module verifies the uploaded image format and prepares it for further processing.

The Image Preprocessing Module standardizes the MRI image by performing grayscale conversion, resizing, normalization, and tensor transformation. These operations ensure compatibility with the trained deep learning models.

The Tumor Segmentation Module employs the Dynamic U-Net architecture to perform pixel-level semantic segmentation of abnormal brain tissues. The model predicts a binary segmentation mask representing the tumor region and overlays the detected area on the original MRI image.

The Explainable AI Module implements Grad-CAM to identify image regions contributing most significantly to the segmentation prediction. The resulting activation map improves model transparency and allows clinicians to verify the decision-making process.

The Radiomics Module extracts quantitative imaging biomarkers from the segmented tumor. These biomarkers describe tumor geometry, intensity distribution, and morphological characteristics, enabling objective characterization of the detected lesion.

The Visualization Module integrates all generated outputs into a single interactive dashboard. Users can simultaneously view the uploaded MRI image, segmented tumor overlay, Grad-CAM heatmap, and radiomic measurements, facilitating comprehensive interpretation of the analysis results.

6. Implementation Details

The proposed system has been implemented entirely in Python using PyTorch as the primary deep learning framework. Dynamic U-Net serves as the main segmentation model responsible for identifying tumor regions within MRI scans, while an additional SegNet architecture is included for experimentation and comparative evaluation. Image preprocessing and visualization operations are performed using OpenCV, Pillow, NumPy, and TorchVision. The user interface is developed using Streamlit, allowing users to upload MRI images and interactively visualize segmentation outputs and Grad-CAM heatmaps. Radiomic features are computed from the segmented tumor regions and stored as structured numerical descriptors, providing quantitative information regarding tumor morphology and intensity. The modular implementation enables each component of the framework to operate independently while maintaining seamless integration within the overall analysis pipeline.

7. Overall Project Workflow

The operational workflow of the proposed system begins with MRI image acquisition through the Streamlit interface. The uploaded image undergoes preprocessing, including grayscale conversion, resizing, normalization, and tensor generation. The processed image is then forwarded to the Dynamic U-Net segmentation model, which produces a binary tumor mask by identifying abnormal tissue regions. The segmented tumor is highlighted on the original MRI image for visual interpretation. Simultaneously, the Grad-CAM module computes attention maps to explain the prediction by identifying the image regions most responsible for the segmentation result. The segmented tumor mask is further processed by the radiomics module to extract quantitative descriptors related to tumor intensity, shape, and volume. Finally, all generated outputs, including the original MRI image, segmented tumor overlay, Grad-CAM heatmap, and radiomic measurements, are displayed through the Streamlit dashboard, providing clinicians with a comprehensive and interpretable analysis of the brain MRI.

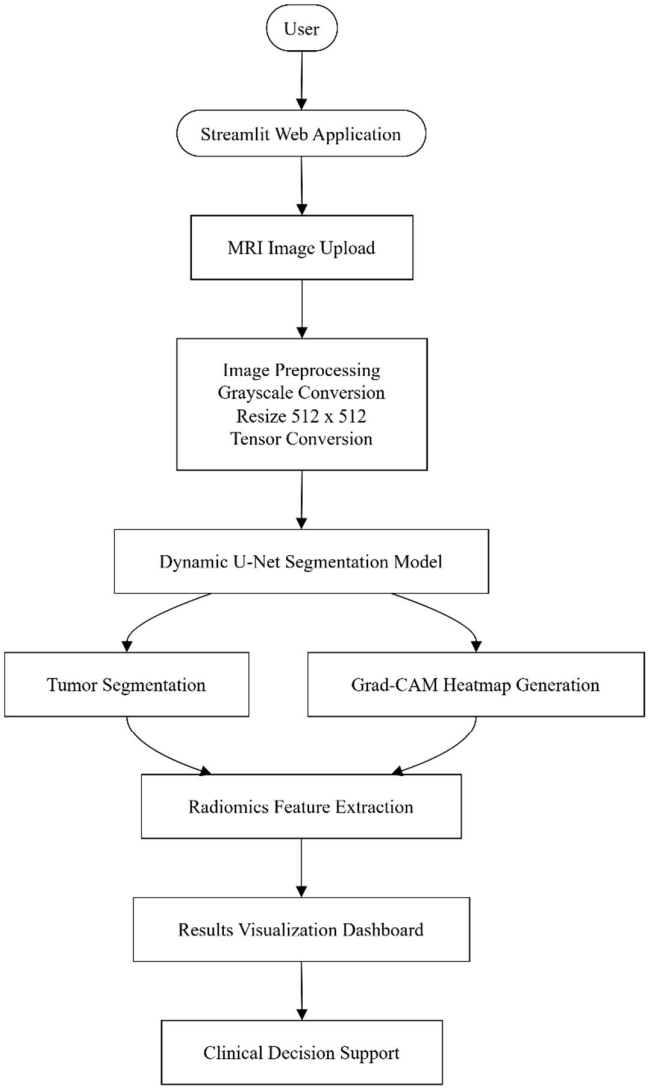


Figure 1. Proposed System Architecture of the AI-Based Brain Tumor Detection and Analysis Framework